

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2022

Math 3113

(Statistics and Numerical Methods)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). An industry has two factories A and B, and the average daily production and standard deviations of the two factories as follows: 12

Factory	Average production	Standard deviation	No. of machine for the production
A	40	5	100
B	80	7	70

(i) In which factory is there greater variation in the distribution of daily production of machine?

(ii) What are the mean and standard deviation of production of all machines of that industry? (Using formula of combined mean and variance).

- 1(b). The following table shows the distribution of the maximum loads in kilonewtons supported by certain cables produced by a company. 23

Determines (i) the mean maximum load, (ii) median of the maximum loads, (iii) mode of the maximum loads, (iv) mean deviation, (v) quartile deviation, (vi) 10-90 percentile range and, (vii) standard deviation.

Maximum load (kN)	No. of cables
93-97	2
98-102	5
103-107	12
108-112	17
113-117	14
118-122	6
123-127	3
128-132	1

- 2(a). What are the postulate of probability? The probabilities that a consumer testing service will rate a new washing machine very poor, poor, fair, good, very good, or excellent are 0.06, 0.13, 0.17, 0.22 and 0.10. What are probabilities that it will rate the new washing machine: (i) very poor, poor, fair, or good; (ii) good, very good, or excellent? 05

- 2(b). Write down five properties of (i) standardized normal probability distribution, (ii) uniform probability distribution, and (iii) exponential probability distribution 12

- 2(c). An industry has two factories A and B with average weekly wages and standard deviation as follows: 10

Factory	Average wages	Standard deviation	No. of workers
A	50	3	15
B	60	12	10

Suppose in factory B, the weekly wages of a worker were wrongly noted as Tk. 80 instead of Tk.40. What is the current mean and variance for the foactory B?

- 2(d). There are 7 types of cement, 5 types of rods, and 8 types of bricks. Find the probabalilty to choose 3 items amount them in which (i) contains cement, rod, and also brick; and (ii) a least any one type of rod is exist. 08

- 3(a). Test whether the following function is probability function (pmf) or not for some value of k: 15

x	-1	0	1	2	3
p(x)	2k	k	0	2k	0.5

Hence, if possible, find (i) $E(5x^2 + 3x)$, (ii) the mean standard deviation, the moment coefficient of skewness and the moment coefficint of kurtosis.

- 3(b). Test whether the following function is probability function (pdf) or not for any value of k: 20

$$f(x) = \begin{cases} ke^{-kx} & ; 0 \leq x \\ 0 & ; \text{otherwise} \end{cases}$$

Hence, if possible final (a) $E(2x^2 + x + 5)$, and (b) mean, standard deviation and the coefficient of skewness. If it is a pdf then, what is the name of this pdf?

- 4(a). What are the conditions that the binomial distribution tends to Poisson distribution? Obtain the mean and variance of Binomial distribution. 15

- 4(b). Three fairs coins are tossed and outputs are displayed as follows: 10

No.of head	0	1	2	3
Observation	6	9	8	5

Find the data to the Poisson distribution. Hence test its goodness of fit with 5% level of significance.

- 4(c). The grade on a class test were 0, 1, 2... 10. The main grade was 5.4 and the standard deviation was 1.5. Assume that the grade is normally distributed. Determine, 10
 (i) the probability of students scoring 6, and
 (ii) the maximum grade of the lowest 20% of the class. Hence, find the estimated number of students scoring 6 among 30 students.

SECTION – B

- 5(a). What is meant by numerical methods? Why should you study them? Mention some numerical techniques for binding roots of nonlinear equation. 10

- 5(b). Narrate absolute, relative and percentage errors. What is meant by root of an equation? Show that the root of the equation $x^2 - 3x + 2 = 0$ is 1.0000 using Newton-Raphson method starting with $x_0 = 0$. 12
- 5(c). Determine the number of men getting wages between Tk.10 and Tk.15 from the following table: 13

Wages in Tk.	No. of men
0-10	9
10-20	30
20-30	35
30-40	42

- 6(a). Find the solution of the following system of linear equation by the Gauss-Seidel method: 13
- $$\begin{aligned} 3x + 2y + z &= 10 \\ 2x + 3y + 2z &= 14 \\ x + 2y + 3z &= 14 \end{aligned}$$

- 6(b). Obtain the approximate value of the root of the equation $3x - \sqrt{(1 + \sin x)} = 0$ by using Bisection method. (Three step only). 10

- 6(c). Find the first, second and third derivatives of the function tabulated below at the point $x = 1.5$: 12

x :	1.5	2.0	2.5	3.0	3.5	4.00
y :	3.375	7.000	13.625	24.000	38.875	59.000

- 7(a). Apply the Lagrange's investigation formula to evaluate the value of y at $x = 0$ for the given set of values $(-2, 5), (1, 7), (3, 11), (7, 34)$. 10

- 7(b). Calculate $\int_2^{10} \frac{dx}{1+x}$ by dividing the range into eight equal parts (up to 4 places of decimal) using Simpson's 1/3 rule. 12

- 7(c). The temperatures T (in $^{\circ}\text{C}$) and length l (in mm) of heated rod are given as : 13

T:	20°	30°	40°	50°	60°	70°
l :	800.3	800.4	800.5	800.7	800.9	801.0

Fit a straight line for the best values of the coefficients regarding T as independent variable.

- 8(a). Solve the system of linear equation by LU decomposition method: 13

$$\begin{aligned} 2x + y + z &= 7 \\ 4x + 2y + 3z &= 4 \\ x - y + z &= 0 \end{aligned}$$

- 8(b). Use Taylor series method to solve the equation 10

$$y' = x^2 + y^2 \text{ for } x = 0.25, \text{ and } x = 0.5 \text{ given that } y(0) = 1$$

- 8(c). Apply fourth order Runge-Kutta method to find an approximate value of y at $x=0.2$ taking $h = 0.1$ for the following differential equation 12

$$\frac{dy}{dx} = x + y^2, y(0) = 2$$

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2022

ESE 3105

(Heat and Mass Transfer) —

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

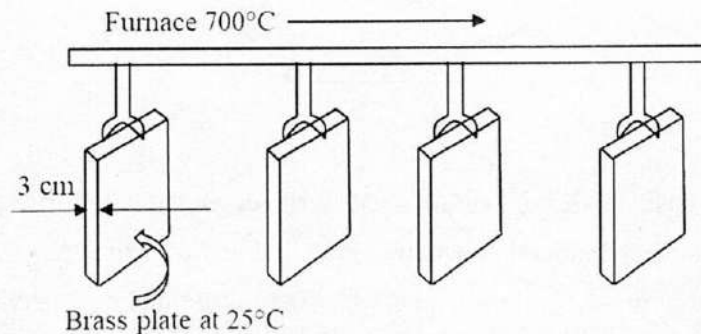
iii) Assume reasonable data if any missing.

iv) Table, chart may be provided on request.

SECTION – A

- 1(a). What are the basic modes of heat transfer? Write down the basic differences among them. 08
- 1(b). Derive general heat conduction equation in cylindrical coordinates. 14
- 1(c). Consider a large plane wall of thickness $L = 0.4 \text{ m}$, thermal conductivity $K = 1.8 \text{ W/m} \cdot \text{K}$, and surface area $A = 30 \text{ m}^2$. The left side of the wall is maintained at a constant temperature of $T_1 = 90^\circ\text{C}$ while the right side loses heat by convection to the surrounding air at $T_\infty = 25^\circ\text{C}$ with a heat transfer coefficient of $h = 24 \text{ W/m}^2 \cdot \text{K}$. Assuming constant thermal conductivity and no heat generation in the wall, determine the rate of heat transfer through the wall. 13
- 2(a). What is “thermal contact resistance”? Write down the name of the factors that affect thermal contact resistance. Define critical thickness of insulation. 10
- 2(b). Suppose a pipe is insulated in such a way that the outer radius of the insulation is less than the critical radius. Now the insulation is taken off. Will the rate of heat transfer from the pipe increase or decrease for the same pipe surface temperature? Justify your answer mathematically. 10
- 2(c). Explain the following term: 08
(i) Efficiency of fin; (ii) Effectiveness of fin.
- 2(d). Consider a 3-m high, 6-m wide, and 0.25-m thick brick wall whose thermal conductivity is $K = 0.8 \text{ W/m} \cdot \text{K}$. On a certain day, the temperatures of the inner and the outer surfaces of the wall measured to be 14°C and 5°C , respectively. Determine the rate of heat loss through the wall on that day. 07
- 3(a). What is lumped system? A spherical copper ball having diameter 12 cm and thermal conductivity $401 \text{ W/m} \cdot \text{K}$ placed in a medium with convection heat transfer coefficient $15 \text{ W/m}^2 \cdot \text{K}$. Will it be appropriate to consider the copper ball as lumped system? Justify your answer. 08
- 3(b). For a lumped system, show that $\frac{\theta}{\theta_i} = e^{-Bi \cdot Fo}$, where the symbols have their usual meanings. 14

- 3(c). In a production facility ^{3-cm}~~3-m~~ thick large brass plates ($K = 110 \text{ W/(m.K)}$, $\rho = 8530 \text{ kg/m}^3$, $C_p = 380 \text{ J/(kg.K)}$, and $\alpha = 33.9 \times 10^{-6} \text{ m}^2/\text{s}$) that are initially at a uniform temperature of 25°C are heated by passing them through an oven maintained at 700°C . The plates remain in the oven for a period of 10 min. Taking the convection heat transfer coefficient to be $h = 80 \text{ W/(m}^2.\text{K)}$, determine the surface temperature of the plates when they come out of the oven. 13



- 4(a). Explain briefly the physical mechanism of boiling with necessary diagram. 10
- 4(b). In case of condensation over a vertical plate, which mode of condensation is more preferable and why? Explain with the help of necessary equations. 07
- 4(c). A large condenser has several column of tube with 33 tubes in each column. The outer diameter of the tube is 1.5 cm . The condenser operates at a pressure of 12.35 kPa and the outer surfaces of the tubes are maintained at 20°C . Determine (i) the average heat transfer coefficient, (ii) the rate of condensation of steam per unit length of a column. 18
- [Following correlation to solve the problem:

$$h = 0.729 \left[\frac{g \rho_l (\rho_l - \rho_v) h_{fg}^* k_l^3}{\mu (T_{sat} - T_s) D} \right]^{1/4}$$

SECTION – B

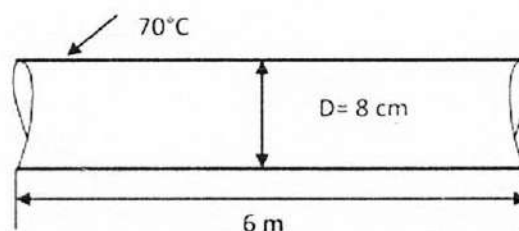
- 5(a). Explain the mechanism of convection heat transfer. What are meant by (i) hydrodynamic boundary layer, and (ii) thermal boundary layer? 11
- 5(b). The thickness of the thermal boundary layer increases in the flow direction, but heat transfer coefficient decreases with the flow direction. Why? Explain. 12
- 5(c). Air flows over a flat plate at a constant velocity of 5 m/s and an atmosphere condition of 1 atm pressure and at a temperature of 28°C , the plate is heated to a constant temperature of 95°C . Starting at a distance of 12 cm from the leading edge, determine the total heat transfer from the plate at a point 40 cm from the leading edge. 12
- 6(a). Derive the integral energy equation for the thermal boundary layer and its thickness. 14
- 6(b). Define film temperature and write down its significance. 06

- 6(c). Some experimental results of a laboratory for the local heat transfer coefficient h_x for flow over a flat plate with an extremely rough surface were found to fit the relation: $h_x(x) = ax^{-0.1}$, where "a" is a coefficient ($W/m^{1.9} \cdot K$) and $x(m)$ is the distance from the leading edge of the plate. Now 15
- Develop an expression for the ratio of the average heat transfer coefficient $\bar{h}_x$ for a plate of length x to the local heat transfer coefficient h_x at x
 - Plot the variation of h_x and $\bar{h}_x$ as a function of x .
- 7(a). What is the concept and application of Reynolds Colburn analogy in connection to heat transfer phenomena? Show the expression of Reynolds Colburn analogy. 10
- 7(b). Write down the differences between forced convection and natural convection of heat transfer with example. 05
- 7(c). Water at $T_1 = 24^\circ C$ is to be heated to $T_2 = 74^\circ C$ by passing it through a tube bundle in staggered tube arrangement. Tubes have an outside diameter 2.5 cm and are maintained at a uniform surface temperature $T_w = 100^\circ C$. The longitudinal and transverse pitches for the staggered arrangement are given by $S_L/D = 1.5$, and $S_T/D = 2.0$. The velocity u_∞ of water just before entering the tube bundle is $u_\infty = 0.3\text{ m/s}$. Calculate (i) the average heat transfer coefficient h_m , (ii) the number of tubes rows in the direction of flow needed to achieve the above temperature size of water. 20
- 8(a). State the Fick's law of diffusion. Write down the differences between absorption and adsorption. 08
- 8(b). What are the different methods of convective heat transfer coefficient evaluation? Discuss the analogy between heat and momentum transfer and the numerical analysis method. 10
- 8(c). A 6 m -long section of a 8 cm diameter horizontal hot water pipe passes through a large room whose temperature is $20^\circ C$ as shown in figure. If the outer surface temperature of the pipe is $70^\circ C$, determine the rate of heat loss from the pipe by natural convection. [Following correlations may be need]. 17

$$Nu = \left\{ 0.6 + \frac{0.387 R_{ad}^{1/6}}{[1 + (0.559/P_r)^{9/16}]^{8/27}} \right\}^2$$

$$Nu = \left\{ 0.825 + \frac{0.387 R_{al}^{1/6}}{[1 + (0.492/P_r)^{9/16}]^{8/27}} \right\}^2$$

$$T_\infty = 20^\circ C$$



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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2022

ESE 3107

(Solar Thermal Engineering)

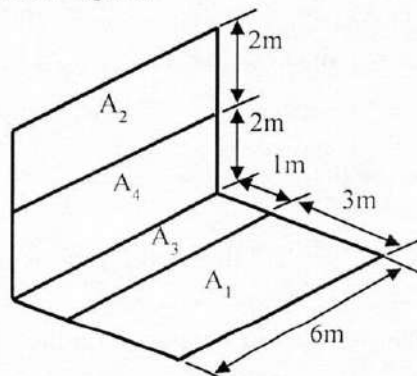
Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.
iv) Table, chart may be provided on request.

SECTION – A

- 1(a). Define the following terms: 15
(i) Air mass, (ii) Hour angle, (iii) Solar azimuth angle, (iv) Beam radiation, and (v) Zenith angle
- 1(b). Derive the following expression of sunshine hour: 10
 $N = \frac{2}{15} \cos^{-1}(-\tan\phi \tan\delta)$, where symbols have their as usual meanings.
- 1(c). Explain the working principle of a Pyrheliometer. 10
- 2(a). What is diffuse fraction? How is daily diffuse fraction related with daily clearness index? 08
- 2(b). What is geometric factor? How to estimate monthly average daily radiation on tilted surfaces? 08
- 2(c). What is black body? State and explain the Stefan-Boltzmann law of radiation. 09
- 2(d). Define the following terms: 10
(i) Emissive power, (ii) Emissivity, (iii) Gray body, (iv) Radiosity, and (v) Summation relationship of view factor.
- 3(a). Derive the expression for net radiation exchange between two infinite long parallel plates using network method. 10
- 3(b). What is reradiating surface? Draw a radiation network for a three-zone enclosure with a reradiating surfaces. 09
- 3(c). Explain the Kirchhoff's law of radiation. 06
- 3(d). Determine the view factor F_{1-2} between two rectangular surfaces A_1 and A_2 for the geometric arrangement shown in the figure. 10



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|-------|--|----|
| 4(a). | What is solar thermal collector? Classfy solar collectors. | 10 |
| 4(b). | Explain the working principle of evacuated tube heat pipe solar collector | 10 |
| 4(c). | Discuss the various components of flat plate solar collector with a neat sketch. | 10 |
| 4(d). | What is the function of glazing materials in solar collector? | 05 |

SECTION – B

- | | | |
|-------|---|----|
| 5(a). | Write down the significance of followings on the performance of a flat plate collector:
(i) Critical radiation level, (ii) Effective transmittance-absorbance product, (iii) Heat removal factor, (iv) Inclination angle. | 08 |
| 5(b). | Derive a general expression to calculate the collector efficiency factor and mention the meaning of each symbols used. | 15 |
| 5(c). | A flat-plate collector of 5 m ² in area is tested to measure the overall heat loss coefficient. Water at 65°C circulates through the collector at a flow rate of 0.05 Lit/sec. The ambient temperature is 8°C and the exit temperature is 50°C. Determine the overall heat loss coefficient. Assume there is no incident solar radiation. | 12 |
| 6(a). | After construction of a new flat-plate solar collector, discuss about the testing procedure to verify the performance. | 10 |
| 6(b). | Define the following terms related to concentrating collector:
(i) Aperture area, (ii) Acceptance angle, (iii) Concentration ratio, (iv) Intercept factor | 10 |
| 6(c). | Schematically show the relationship between concentration ratio and temperature of receivers. Also, discuss the practical ranges of concentration ratios for the operation of various types of concentrating collectors. | 10 |
| 6(d). | What characteristics a reflector should be possessed? | 05 |
| 7(a). | Schematically show and discuss the solar heat pump systems for (i) water heating, and (ii) space heating. | 12 |
| 7(b). | What are the advantages and disadvantages of passive solar water-heating system? | 08 |
| 7(c). | How to prevent reverse circulation in passive solar water heating systems. | 05 |
| 7(d). | A fully mixed water storage tank contains 400 kg water, and has a UA product of 10 W/°C , and is located in a room that is at constant temperature of 16°C. The tank is examined in a 8 h period starting from 7 AM where Q _u is equal to 0, 8, 20, 28, 40, 55, 65, 55 MJ. The load is constant and equal to 10 MJ in first 2h, 14 MJ in the next 2h, 18 MJ in the next 2h, and 24 MJ the rest of the time. Find the final storage tank temperature if initial temperatures is 40°C. | 10 |
| 8(a). | What is solar crop-drying? Describe the working procedure of an indirect solar crop-drying system with schematic diagram. | 13 |
| 8(b). | With the help of necessary diagram, explain the working principle of solar assisted liquid desiccant cooling system. | 10 |
| 8(c). | What are solar pond and solar thermal power system? With the schematic diagram, explain the solar power tower system to generate electricity | 12 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2022

EE 3113

(Power System Engineering)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

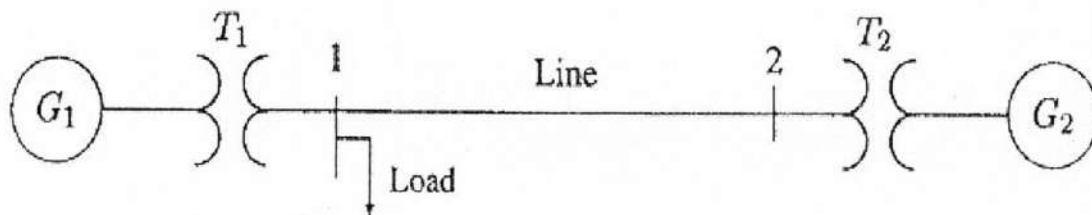
iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Draw the one line diagram of a typical a.c power supply scheme, and explain briefly different sections of it. 10
- 1(b). What is the purpose of an overhead transmission line? “The present trend in towards a.c. for generation and distribution, and d.c. for transmission”. Discuss the reasons for it. 10
- 1(c). What are the advantages of using ACSR conductors? Determine the inductance of a two wire transmission line due to internal and external flux. 15
- 2(a). “The greater the transmission voltage, the lesser is the conductor material required” Justify the statement with required derivation. How can transmission efficiency be increased with higher transmission voltage? 15
- 2(b). What is meant by electric potential? Derive an expression for the capacitance of a single phase overhead transmission line. 10
- 2(c). A 3-phase, 50Hz, 150 km line has a resistance, inductive reactance and capacitive shunt admittance of 0.1Ω , 0.5Ω and $3 \times 10^{-6} \text{ S}$ per km per phase. If the line delivers 50 MW at 110 kV and 0.8 p.f. lagging, determine the sending end voltage and current. Assume a nominal π circuit for the line. 10
- 3(a). Using rigorous method, derive expressions for sending end voltage and current for a long transmission line. 15
- 3(b). Describe the various methods for reducing corona effect in an overhead transmission line. 10
- 3(c). An overhead transmission line conductor having a parabolic configuration weighs 1.925 kg per metre of length. The area of X-section of the conductor is 2.2 cm^2 and the ultimate strength is 8000 kg/cm^2 . The supports are 600 m apart having 15 m difference of levels. Calculate the sag from the taller of the two supports which must be allowed so that the factor of safety shall be 5. Assume that ice load is 1 kg per metre run and there is no wind pressure. 10
- 4(a). What is meant by string efficiency? Determine the string efficiency of a insulator string composed of 3 identical suspension insulators. Also explain the methods of improving string efficiency. 15
- 4(b). What are the the desirable properties of line supports and insulators? 10
- 4(c). An insulator string consists of three units, each having a safe working voltage of 15 kV. The ratio of self-capacitance to shunt capacitance of each unit is 8:1. Find the maximum safe working voltage of the string. Also find the string efficiency. 10

SECTION – B

- 5(a). What is P.U. system? Write the benefits of P.U. system for analyzing the power system. 10
- 5(b). Show that the equivalent impedance of a transformer in per unit referred to the LV and HV side is same. 10
- 5(c). Draw an impedance diagram for the electric power system shown in figure. Showing all impedances in per unit on a 100 MVA base. Choose 20kV as the voltage base for generator. The three-phase power and line-line ratings are given below. 15



G_1 :	90 MVA	20 kV	$X = 9\%$
T_1 :	80 MVA	20/200 kV	$X = 16\%$
T_2 :	80 MVA	200/20 kV	$X = 20\%$
G_2 :	90 MVA	18 kV	$X = 9\%$
Line :		200 kV	$X = 120 \Omega$
Load :		200 kV	$S = 48 \text{ MW} + j 64 \text{ Mvar}$

- 6(a). Why are isolators not opened on load? Describe with the aid of neat sketch the construction and working of a Tirril regulator, for voltage control in power system. 14
- 6(b). A fuse wire of circular cross-section has a radius of 0.8 mm. The wire blows off at a current of 8A. Calculate the radius of the wire that will blow off at a current of 1A. 05
- 6(c). Discuss the fundamental requirements of protective relaying. 07
- 6(d). Name the factors that should be taken care of while designing and erecting a substation. Draw the key diagram of a typical 11 kV/400 V indoor sub-station. 09
- 7(a). What is voltage source? Discuss the causes of over voltages. 10
- 7(b). Why is ground wire used in equipment grounding? Write a short note about ungrounded enclosure and enclosure connected to neutral wire for equipment grounding. 10
- 7(c). Why is current interruption easier in an a.c. circuit than in a d.c. system? 04
- 7(d). Discuss the principle of operation of an axial-blast air circuit breaker. What are the advantages and disadvantages of using air as the arc quenching medium? 11
- 8(a). What is meant by sequence networks? Derive an expression for fault current of line-to-line fault by symmetrical components methods and draw the sequence network connection for the line-to-line fault. 15
- 8(b). What is the reason of transmission interconnection? What is the impact of cyber security on the smart grid? 10
- 8(c). Write the difference between conventional grid and smart grid in power system. Discuss the key functions of smart grid with smart grid's communication system. 10

-End-

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2022

ESE 3123

(Thermo-Fluid Devices)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

iv) Table, chart may be provided on request.

SECTION – A

- 1(a). Define hydraulic diameter. Show the different flow regions of a fluid flowing through a pipe and demonstrate how it differs with Reynolds number? 10
- 1(b). What are the different types of losses when a fluid flowing through a pipe? Derive the expression for the loss of head due to friction in pipes. 15
- 1(c). A submarine "Titan" moves horizontally in sea and has its axis 15 m below the surface of water. A pitot-tube properly placed just in front of the submarine "Titan" and along its axis is connected to the two limbs of a U-tube containing mercury. The difference of mercury level is found to be 190 mm. Find the speed of the submarine knowing that the specific gravity of mercury is 13.6 and that of sea-water is 1.026 with respect of fresh water. 10
- 2(a). What is meant by dimensional homogeneity? Briefly discuss about it with an example. 06
- 2(b). What are the different flow rate measurement instrument that are commonly use? Write down the differences between Venturimeter and Orifice meter. 06
- 2(c). Derive an expression for the volumetric flow rate of a fluid flowing through a nozzle meter. 12
- 2(d). A horizontal venturimeter with inlet and throat diameters 30 cm and 15 cm, respectively, is used to measure the flow of water. The reading of differential manometer connected to the inlet and the throat is 20 cm of mercury. Determine the rate of flow. Take $C_d = 0.98$. 11
- 3(a). Define hydraulic machines? Explain the working principle of a reciprocating pump. 10
- 3(b). Write short notes on: (i) Manometric efficiency, (ii) Static head, (iii) Mechanical efficiency, and (iv) Manometric head. 10
- 3(c). Describe the casing with guide blade of a centrifugal pump with a schematic diagram. 10
- 3(d). Write down the basic differences between fan, compressor and blower. 05
- 4(a). What is meant by specific speed of a pump? Show that, the specific speed is dependent to the manometric head and the rate of discharge. 12
- 4(b). What is the significance of characteristics curve of a centrifugal pump? Discuss the main and operating characteristic curves. 08
- 4(c). Define cavitation. Why and where it occurred? Also write down the significance of NPSH in pump industry. 07
- 4(d). A double-stage centrifugal pump with impeller diameter of 30 cm rotates at 2000 rpm and lifts 5 m^3 of water per second to a height of 25 m with an efficiency of 80%. Find the number of stage and diameter of each impeller of a similar multistage pump to lift 7 m^3 of water per second to a height of 200 m when rotating at 1500 rpm. 08

SECTION – B

- 5(a). Write down the differences between compressible and incompressible flow. Write down the classification of compressible flow according to the variation of Mach number. 11
- 5(b). Discuss the effect of back pressure on the mass flow rate and the pressure distribution along converging nozzle with neat sketch. 12
- 5(c). Air enters a nozzle at 0.2 MPa, 350K, and a velocity of 150 m/s. Approximating the flow as isentropic, determine the pressure and temperature of air at a location where the air velocity equals the speed of sound. What is the ratio of area at this location to the area of entrance? 12
- 6(a). Write short notes on the followings: 06
(i) Shock wave, (ii) Normal shock wave, and (iii) Oblique shock wave.
- 6(b). Show that the relationship between Mach numbers upstream and downstream of normal shock wave can be given by: 17
- $$\left(M_{a_2}^2 = \frac{M_{a_1}^2 + \frac{2}{k-1}}{\frac{2M_{a_1}^2 k}{k-1} - 1} \right) \text{ where the symbols have their usual meanings.}$$
- 6(c). Air flowing steadily in a nozzle experiences a normal shock at a Mach number of $M_a = 2.5$. If the pressure and temperature of air are 61.64 kPa and 262.15K, respectively at upstream of the shock, calculate the pressure, temperature, velocity, Mach number, and stagnation pressure at downstream of the shock. 12
- 7(a). Write down the classification of heat exchanger. Also write short note on the fouling factor. 08
- 7(b). Derive an expression of LMTD for counter flow heat exchanger. 14
- 7(c). Water at the rate of 68 kg/min is heated from 35 to 75°C by an oil having a specific heat of 1.9kJ/(kg. °C). The fluids are used in a counter flow double-pipe heat exchanger, and the oil enters the heat exchanger at 110°C and leaves at 75°C. The overall heat transfer coefficient is 320 W/(m². °C). Calculate the heat exchange area. 13
- 8(a). What is meant by effectiveness of heat exchanger? Write down its significance. 06
- 8(b). Justify the statement "A counter flow heat exchanger has higher effectiveness than a parallel flow heat exchanger at the same condition". 06
- 8(c). Derive the expression for overall heat transfer coefficient for a heat exchanger by considering the fouling factor. 11
- 8(d). A counter flow heat exchanger of heat transfer area $A = 12.5 \text{ m}^2$ is to cool oil [$c_{ph} = 2000 \text{ J/(kg. °C)}$] with water [$c_{pc} = 4170 \text{ J/(kg. °C)}$]. The oil enters at $T_{h,in} = 100 \text{ °C}$ and $m_h = 2 \text{ kg/s}$, while the water enters at $T_{c,in} = 20 \text{ °C}$ and $m_c = 0.48 \text{ kg/s}$. The overall heat transfers coefficient is $U_m = 400 \text{ W/(m}^2 \cdot \text{°C)}$. Calculate the exit temperature of water $T_{c,out}$ and the total heat transfer rate Q . 12

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